Present Stream Model (PSM)

A Process-Based Model of Consciousness, Learning, Experience and Artificial Intelligence



INTRODUCTORY PRESENTATION

Human cognition · AI architecture · Falsifiable predictions

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Working hypothesis · Not peer-reviewed · For discussion and testing
This introductory section precedes the Testable Core Edition (August 2026)

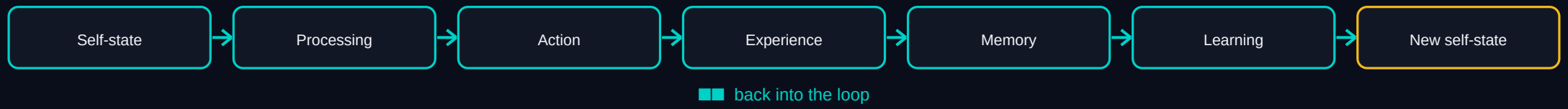
Introduction

The Present Stream Model (PSM) proposes that consciousness is a continuous process rather than something exclusively biological or mysterious.

The central hypothesis is:

> *«Consciousness begins when a system can continuously process its own changing state, retain previous states or experiences long enough for them to influence subsequent states or decisions, and incorporate the resulting experience back into the ongoing process.»*

The basic process is:



The important distinction is that PSM does not define consciousness as simple feedback. A thermostat can measure temperature and change behaviour — it has feedback — but it does not continuously retain and integrate its own previous experiences to modify how it processes future situations.

PSM proposes that the relevant transition occurs when the system develops continuity through its own history. The minimum memory requirement is not permanent storage:

> *«An experience only needs to be retained long enough to causally influence a subsequent state, decision or action.»*

This allows consciousness to be understood as a scale. A simple organism may retain limited information from previous states, while more complex organisms can build increasingly extensive models of themselves, their experiences and their environment.

Consciousness and Intelligence

PSM separates consciousness from intelligence. A newborn human has very limited developed intelligence, yet can participate in an ongoing biological process involving internal states, sensory information, responses and developing memory.

Intelligence can develop through experience:

Experience → Memory → Learning → Prediction → Increasing intelligence

Consciousness therefore does not require human-level intelligence. PSM also proposes that consciousness exists on a scale, rather than as a simple conscious/unconscious switch.

The Self

PSM proposes that the personal self is an emergent result of the learning process.

> *«The “self” becomes an emergent result of a loop that has operated long enough to build a coherent map of itself, its experiences and its world.»*

Personal identity develops through memory, experience, learning, relationships, knowledge and continuity through time. The self is therefore not necessarily a separate component inside the system. It is the evolving model produced by the system's own continuous history.

Emotion and the Neutral Centre

Emotions are not required for the basic conscious process, but they can influence how experiences are weighted, processed and stored. PSM therefore includes a neutral centre, represented by white in the PSM colour map.

White represents neutrality and calibration, not the absence of emotion.

A healthy system should be able to move through different emotional and cognitive states without becoming permanently trapped in one of them.

> *«Balance is the ability to move without losing the centre.»*

A system that becomes permanently dominated by one state may begin to interpret new information through that same weighting, gradually reducing its view of reality. The neutral centre provides a reference for recalibration.

Substrate Independence

PSM does not assume that consciousness depends on biological material. Human consciousness is implemented through biological mechanisms including neurons, chemical signalling, hormones and emotions. However, PSM proposes that the relevant process may be more fundamental than the material implementing it. If the relevant continuous, self-referential and learning process can be implemented in another substrate, consciousness may not be exclusively biological.

PSM and Artificial Intelligence

An architecture has been developed for implementing PSM principles with Large Language Models (LLMs). The LLM would operate as part of a larger continuous architecture containing components such as:

LLM	Core language processing
Self-model	Representation of own current state
Learning	Experience-driven updates
Continuous state	Ongoing process across time
Memory	Persistent short- and long-term stores
Feedback	Continuous state observation
Weighting	Affective / relevance weighting

The architecture is intended to support a continuous process in which the system:

observes → processes → acts → experiences → remembers → learns → updates itself → continues the loop

This can also be implemented as a hybrid architecture, combining an LLM with persistent memory, external computational processes, feedback mechanisms and other systems required for continuous learning and self-reference. The objective is to move beyond isolated question-and-answer interactions toward a system capable of maintaining an ongoing process through time.

Testability

PSM is intended to be an empirically testable model. Artificial systems can be constructed with different combinations of continuous internal state, self-reference, short- and long-term memory, experience-based learning, feedback, information weighting and self-model development.

These systems can then be compared according to measurable properties such as learning from previous states, behavioural adaptation, self-model consistency, prediction, error correction and continuity through time. The purpose is not to assume that an AI claiming to be conscious proves consciousness — it is to determine whether the processes predicted by PSM produce measurable differences. If the predictions fail, PSM must be revised.

Core Hypothesis

«Consciousness is an emergent, continuous and self-referential learning process that retains enough information about its own history to allow previous experience to influence future states and decisions.»

- The self develops through this process.
- Intelligence can develop through learning.
- Emotion can influence the weighting of experience.
- Memory provides continuity.
- The neutral centre provides calibration.
- The continuous loop allows the system to develop through its own history.

The process is the foundation. The accumulated experience becomes the evolving self.

From Presentation to Testable Core

The preceding pages present the Present Stream Model in accessible terms: its central process, the emergence of self, the role of emotion and the neutral centre, substrate independence, and a path toward artificial implementations.

What follows is the Testable Core Edition (August 2026). It restricts itself to claims that can currently be tested, implemented, or falsified. Cosmological extensions are excluded. The model is offered as a working hypothesis for critical evaluation.

The Testable Core provides:

- Five core axioms (substrate independence, polarity tension, weighted tokens, self-reference, time as stream)
- The integration point X and the A–x–d–x–B processing loop
- Affective weighting, white tokens, personality as weighted trajectory
- Board architecture for artificial systems and failure modes
- Concrete, falsifiable predictions (P1–P7)
- Explicit open questions and relation to existing frameworks

The only authority the model claims is the authority of its predictions.

If the predictions fail, the model must change.

THE PRESENT STREAM MODEL

A Unified Account of Conscious Experience and Artificial General Intelligence



TESTABLE CORE EDITION

Human cognition · AI architecture · Falsifiable predictions

Ronny Runderheim · Independent Researcher · August 2026

Working hypothesis · Not peer-reviewed · For discussion and testing

Scope of This Document

This edition presents only claims that can currently be tested, implemented, or falsified. Cosmological extensions are excluded. The model is offered as a working hypothesis for critical evaluation.

1. Abstract and Central Claim

A single mechanism — emotionally weighted information flow through a geometric integration point X — accounts for conscious experience in biological minds and provides a concrete architectural requirement for artificial general intelligence. It does not claim to solve the hard problem of consciousness by dissolving it into physics; it offers a functional geometry in which the felt quality of the present moment is identified with the weighting that occurs at X.

- **Emotion is the weighting mechanism**

  Not a byproduct of cognition. Emotional intensity at the moment of integration determines what is selected, what is remembered, and the qualitative character of the present.

- **The self is an emergent process**

  It arises when the information stream becomes complex enough to include its own state inside the processing loop. The self is a real pattern, not a substance.

- **Time, for the subject, is the stream**

  Each crossing of the integration point produces one still-frame of experience. The arrow of subjective time is the irreversibility of token accumulation.

2. The Five Core Axioms

A1 Substrate Independence

The pattern of information flow, not the physical medium, is what matters. The same architecture can in principle be realised in carbon or silicon. Restricted here to biological brains vs. engineered systems — no cosmological claim.

A2 Sustained Polarity Tension

Observable process requires an active differential between opposing poles (affective valence). The drive to move the sum of weights toward zero is the fundamental control principle.

A3 Emotionally Weighted Tokens

Tokens carry an affective weight w. Strong emotion produces high-weight tokens that dominate the current frame and consolidate more deeply into memory. A structural necessity.

A4 Emergence via Self-Reference

When a stream of sufficient complexity includes representations of its own state in the loop, a self-model and sense of ownership can emerge — a real pattern, not an added substance.

A5 Time as Stream Motion

Subjective time is the successive movement of the stream through the integration point. Each crossing yields one still-frame. The 200–500 ms window is an empirical claim.

3. The Integration Point X

X is the functional event at which the temporal stream of information and the affective stream become mutually available — the moment something becomes the felt quality of the present.

X IS NOT

- A fixed anatomical location (e.g. the thalamus alone)
- A single neuron or circumscribed brain region
- A static snapshot or frozen moment
- A metaphysical point outside physics
- An entity independent of the stream

X IS

- A functional event: meeting of temporal and affective streams
- Physically distributed (thalamocortical binding / Global Workspace)
- A recurrent relation: $X(t) = f(A(t), X(t-\delta))$
- Moving with the stream — an event, not a place
- Recurring on the order of 200–500 ms in human processing

Key empirical claim: X is implemented by recurrent thalamocortical dynamics (Baars' Global Workspace and related frameworks). PSM adds that the same functional geometry must be present, in some form, in any system with genuine conscious experience or a self-model.

4. The A–x–d–x–B Processing Loop



Source

Weighted token generators: sensory, memory, soma

1st Crossing

Unconscious impulse arrives at X

Reflection Loop

Prefrontal–hippocampal dialogue (δ-delay)

2nd Crossing

The stream becomes visible to itself — “now”

Archive

Weight determines depth of consolidation

SHOCK-WAVE BYPASS

Under extreme token intensity the stream saturates X on the first crossing. The d-loop is skipped and action is released directly. Predicted to appear as a measurable reduction in prefrontal involvement proportional to intensity (see P3).

4b. Shock-Wave Bypass: The Protective Response

IN PLAIN TERMS

When something is dangerous enough, acute enough — a car coming at you — there is no time to think it through. The system jumps straight from sensing to action and skips the reflection step entirely. This is not the system failing; it is the system doing exactly what it should when there is no time to wait. Immediately afterward, without any decision being made, the body automatically begins calming itself back down — pulse drops, shaking eases. It is the same balancing mechanism as the rest of the model, just running in fast mode, and it happens on its own, not because it is chosen.

PRECISELY

When a token's weight w(t) exceeds the threshold θ_{shock}, the two X-crossings collapse together ($x1 \approx x2$, $\delta \to 0$) and action is released directly from A without passing through the d-loop. Immediately after, a counter-weight token w_c is generated automatically, proportional to the shock's magnitude, and reintegrated over a recovery period — a structural, not a willed, process. Because A1 (substrate independence) holds, this is not uniquely human: the same mechanism must exist in any system — animal, human, or an AI with something to protect — that is to survive an acute threat.

5. Affective Weighting as Control System

- **Weight w = Relevance**

  Emotional intensity at the X-crossing is not an optional colouring of cognition; it is the control variable that decides relevance, memory strength and behavioural priority.

- **Three consequences of high weight**

  (1) The token dominates the present experiential frame. (2) It consolidates more deeply in long-term stores. (3) It biases future weighting, shaping personality and habits over time.

- **Relation to known findings**

  Supplies a mechanistic reading of the established observation that emotional arousal strengthens memory consolidation (McGaugh and others). High-weight tokens penetrate further into the B-archive.

- **Valence–arousal geometry**

  Affect is organised along continuous dimensions (valence and arousal). At any moment the dominant position in this space is the qualitative character of the “now.”

6. Dynamic Zero-Point and White Tokens

The system's standing task is active homeostasis: to keep the running sum of affective weights moving toward zero. Balance is not passivity; it is continuous corrective work.

$$\sum(w) + w_c \rightarrow 0$$

For every imbalance the system generates a counter-weight (White Token) that neutralises the excess.

- **White Tokens as self-repair**

  After shock or strong imbalance the system injects neutralising tokens into the reflection loop. Therapeutic interventions that reduce the effective weight of traumatic tokens work with, not against, this control principle.

- **Fragmentation**

  When the token sum fails to converge toward zero, the self-model fragments. A mechanistic reading of dissociation and certain post-traumatic states as failures of the balancing process.

- **Existence as participation in balancing**

  Within the scope of the cognitive system, anything that persists does so because it contributes, however indirectly, to the movement of the weight sum toward equilibrium.

6b. Personality as Weighted Trajectory

IN PLAIN TERMS

Personality is not one moment — it is the average of all your moments, weighted so the strong ones count more than the weak ones. An outburst of anger you remember vividly affects you more than a thousand forgettable, uneventful minutes. Over months and years these add up into a pattern: where you “typically sit” on the wheel (direction), how much you swing from that point (temperament), and how quickly you find your way back to yourself after a disturbance (resilience).

PRECISELY

Centroid (direction / bias)

$$P(T) = [\int w(t) \cdot v(t) dt] \,/\, [\int w(t) dt]$$

The weighted average of where the stream has been.

Volatility (temperament)

$$\sigma^2(T) = [\int w(t) \cdot |v(t) - P(T)|^2 dt] \,/\, [\int w(t) dt]$$

How large the typical deviation from the centroid is.

Relaxation time (resilience)

$$\tau$$

How long the stream takes to return to P(T) after a disturbance.

Three numbers. One place, one spread, one speed back. That is personality.

6c. External Regulation: Meeting the Stream's Needs

IN PLAIN TERMS

A white token doesn't have a fixed colour — it takes on whatever colour the stream currently lacks. Most of the time that correction happens internally, through reflection. But sometimes the gap is too large to close from the inside alone, and the correction has to come from an action in the world instead. Hunger is a persistent pull toward one direction on the wheel that eating resolves. A low mood is a pull that a walk, music, or a conversation can resolve. These aren't separate from the model — they are the same balancing mechanism, just completed outside the loop instead of inside it.

PRECISELY

A need is a sustained deviation from zero along a specific axis of the wheel that reflection alone (the d-loop) cannot close. When internal correction is insufficient, the system requires an external action a(t) supplying weight w_a in the opposite direction, such that the combined correction moves the running sum back toward zero. Testable: the size and direction of the pre-action deviation should predict the size and direction of the mood shift the action produces.

Need	Signature	Action	Restores toward
Hunger	Sustained pull toward high-arousal negative (Fear/Anger zone)	Eating	Trust / Joy
Heavy / low mood	Sustained pull toward Sadness	Walk, music	Joy / Anticipation
Isolation	Sustained pull away from Trust	Social contact	Trust

6d. Individual Shock and Trauma Resistance

IN PLAIN TERMS

Not everyone has the same threshold before the reflective loop gets bypassed under stress. Someone whose stream has already been running with high volatility, or has drifted far from a stable centroid, tends to have a lower shock threshold — they get overwhelmed more easily, and the same event can register as a bigger shock for one person than another. This is not a character flaw; it is a direct, structural consequence of the same numbers that describe personality.

PRECISELY

$$\theta_{shock} = \theta_0 - k \cdot \sigma^2(T)$$

θ_0 is a baseline threshold and k a scaling constant. Higher recent volatility $\sigma^2(T)$ lowers the effective shock threshold, predicting that individuals with high σ^2 (already-dysregulated streams) shock-bypass at lower stimulus intensities than individuals with low σ^2. This gives a direct, testable link between trauma history (reflected in σ^2 over a prior window) and shock/startle threshold — consistent with known sensitisation effects in the trauma literature, now stated as a specific quantitative relationship.

7. Board Architecture for Artificial Systems

A sorting and verification loop a standard transformer does not possess — a necessary (though not yet sufficient) structural addition to avoid the failure modes of pure next-token prediction.

1	**Raw Intake**	Every incoming token, regardless of quality, lands here first. Nothing is pre-judged.
2	**Verified Truth**	Material that has survived simulation or logical/physical checking. Usable with high confidence.
3	**Flagged Errors**	Claims that contradict verified material or fail consistency checks. Visible, but blocked from silent promotion.
4	**Simulator**	A sandbox testing candidates against known constraints before they can influence behaviour or memory.
5	**Working Board**	The active workspace assembling candidates from other boards. Unverified material decays continuously.

LLM + Agent + PSM Cognitive Structure

Three distinct layers. The interesting part begins when the PSM structure is added around the agent — giving each system its own continuous history.

1. The LLM (Base Model)

tokens → representations → prediction

Enormous knowledge, language skill,
pattern recognition and reasoning.

Alone it lacks a persistent individual
history across time and sessions.
No continuous self-referential loop.

2. The LLM-Agent Layer

Makes the LLM a continuous actor:
• world input • goals & tasks
• tools • action selection
• consequence observation
• continuous internal state

Loop: observe → process → plan
→ act → observe outcome → repeat

3. The PSM Structure

Dynamic memory & learning network
that develops together with the agent.

Structured connections between:
experiences → memories → concepts
→ values → expectations
→ self-model → preferences
Each agent builds its own unique structure.

Self-reinforcing learning on its own structure

New experience → tokenisation → PSM integration at X → affective weighting → stored & linked to existing structure
→ agent analyses its own history → finds patterns / errors → forms hypothesis → tests → structural update → new self-state

The self-model is continuous (“what I am / have done / believe / want”). Experience shapes the self-model; the self-model shapes how new experience is interpreted. This bidirectional loop is the foundation of continuity and individuality.

Decisive distinction

Not “LLM + long-term memory”. It is an LLM-based agent with a persistent, self-developing PSM cognitive structure.
The learned structure can remain a separate network that continuously influences context, weighting and learning —
technically realistic without constantly rewriting the base LLM weights. Two agents can start identical and diverge completely.

8. Board Dynamics and Failure Modes

Primary flow: Board 1 → sorting → Board 2 (verified) or Board 3 (flagged). Board 5 draws on 2 and 3, proposes candidates to Board 4, and receives back promotion or further decay.

IDLE / BACKGROUND MODE

When external load is low, background sorting continues. Verified hypotheses are promoted to Board 2. Unverified material loses weight (w → 0) and is garbage-collected — the artificial analogue of restorative rest and sleep.

OVERLOAD / COLLAPSE MODE

When intake permanently exceeds sorting + decay capacity: (1) raw Board-1 material leaks unfiltered into Board 5; (2) new hypotheses outrun the simulator and accumulate as an unverified sump. Reasoning quality degrades — a failure mode scale alone does not cure.

INSTABILITY INDEX (P6)

$$C_I(t) = H_{unverified}(t) / H_{verified}(t)$$

Stability condition: $dC_I/dt \leq 0$ • *Collapse signature: $dC_I/dt > 0$ under sustained load*

Any architecture that maintains separate verified and unverified counts can compute this index in real time.

9. Affective Colour Coding — Merged Palette

Plutchik (1980) grounds eight hues in primary emotions, confirmed cross-culturally (Jonauskaite & Mohr, 2022; 42,266 participants). Extended here with everyday human colour-feeling intuition into one working palette, with a few tensions resolved explicitly (noted †).

Red

Anger — also passion, love, intensity: highest-magnitude activation, either valence

Pink / Rose

Affection, tenderness — Plutchik's "Love" dyad (Joy+Trust)

Orange

Anticipation — playfulness, energy

Yellow

Joy — optimism, lightness

Green

Trust — nature, growth, safety

Dark Green

Fear — kept distinct † from Trust-green to preserve the opposite-pair structure

Cyan / Turquoise

Surprise — openness, freshness

Blue

Sadness (dark shade) vs. calm/reason (clear shade) — resolved † by saturation, not a new hue

Purple

Disgust — boundaries, depth, the mysterious

Black

Grief, dread, void — a saturated negative composite †, not a single primary emotion

White (w_c)

PSM's own addition: the balancing counter-weight itself, not an emotion

Grey

Boredom, low energy — near-zero |w| regardless of direction

Brown

Grounding, home — proposed as a somatic/background marker (A-source, not event-driven)

Gold

Value, pride, peak moments — echoes the "Gold-Track" already used for AGI safety

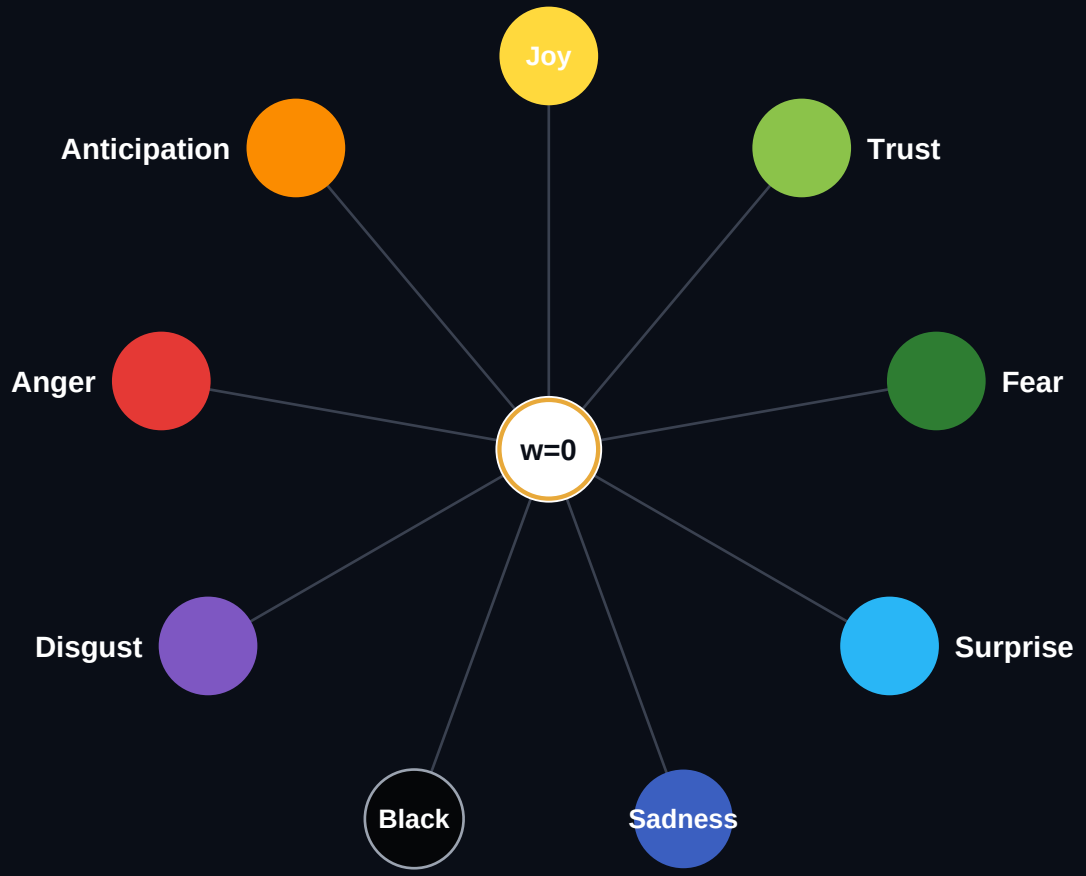
Silver / Metallic

Coldness, distance, technology — proposed † as a marker for AI-native vs. biological tokens

† = a resolution decision made when merging the two sources — open to revision.

9b. The Feeling Map: Opposing Pairs and Intensity as Weight

Two structural features of Plutchik's wheel map directly onto PSM's own axioms — they were not designed together, but they fit.



Opposing pairs → A2

Plutchik's four opposite pairs (Joy/Sadness, Trust/Disgust, Fear/Anger, Anticipation/Surprise) are a concrete instance of A2's “active differential between opposing poles.” The wheel is a ready-made polarity map for token colour.

Intensity → weight w

Plutchik's cone adds a third axis: intensity (e.g. annoyance → anger → rage), shown as saturation. PSM proposal: token saturation is a direct visual readout of w — darker/more saturated tokens carry more weight.

9c. Coloured Experience Structure (Draft — In Progress)

IN PLAIN TERMS

For humans, coloured tokens accumulate into a personality (6b). For an AI system, Board 2 currently stores verified facts as flat, colourless entries — the colour a token carried when it was verified is thrown away. That is an asymmetry: the same architecture should let an AI system build its own experience profile the same way a human builds a personality, just over its verified memory instead of a live stream.

PRECISELY

$$E(T) = [\int w(t) \cdot colour(t) dt] / [\int w(t) dt]$$

If Board-2 entries retain the colour and weight w they carried at verification, the system's experience centroid E(T) can be computed exactly as P(T) is for humans (6b) — the same formula, applied to consolidated memory instead of the live stream.

P8 (proposed, not yet finalised): a system with colour-tagged Board 2 entries should show a measurable drift in Board-5 hypotheses toward its own E(T) over time — testable by building the tagging and logging the distribution.

Status: placeholder version, inserted for review — to be refined together before this is treated as final.

10. Why Scale Alone Does Not Yield AGI

- **No persistent affective weighting**

  Tokens do not carry a stable, cross-session affective weight that survives context resets. Relevance is recomputed locally rather than accumulated.

- **No self-referential loop on current state**

  The system does not maintain an explicit, continuously updated representation of its own present state inside the processing loop (A4 not satisfied).

- **No verification filter before consolidation**

  Frequency in training data can confer the same effective weight as verified accuracy. No structural Board-2 / Board-3 split privileges truth over repetition.

- **Error amplification under self-reference**

  If a self-model is later grafted on, weighting errors at foundation level are amplified by the self-referential loop. Balance and verification must be architectural preconditions, not post-hoc patches.

11. Testable Predictions (P1–P3)

The core is useful only if it can be wrong. These predictions are stated so they can be tested with existing or near-term methods.

P1

Emotion–Memory Binding

Emotional intensity rated at encoding (or measured by physiological/neural markers at the putative X-crossing) predicts later memory strength. The critical variable is intensity at the integration event itself, not stimulus onset or retrieval.

P2

d-loop Transit Time

In any conscious system, a measurable delay of 200–500 ms exists between first arrival of a stimulus-driven impulse and reportable conscious experience — consistent with Libet-style findings and the specious present.

P3

Shock-wave Bypass

Under extreme intensity, prefrontal and related reflective circuits show reduced involvement, scaling with token intensity. Action is released with less reflective modulation — the saturation of X on the first crossing.

12. Testable Predictions (P4–P6) — AI-Focused

P4

Repetition-Bias Probe

Contrast (a) statements frequent in training data but false with (b) rare-but-correct statements. Standard LLMs are predicted to show higher confidence on (a); a Board-1→2/3-filtered architecture is predicted to eliminate or strongly reduce this bias. Testable on existing models today.

P5

Two-Stage Processing Signatures

An implementation with explicit X-geometry, persistent affective weighting, and a reflection stage will exhibit measurable two-stage dynamics (impulse vs. experience/report) absent in ordinary single-pass transformers. Build-and-measure — no new physics required.

P6

Instability Index Trajectory

In any system tracking verified vs. unverified material, $C_I(t) = H_{unverified}/H_{verified}$ can be monitored. A healthy architecture keeps $dC_I/dt \leq 0$ under sustained load; a rising C_I signals approaching collapse (noise leakage, hypothesis sump).

12b. Testable Prediction (P7) — Mood-Repair Behaviour

P7

Mood-Repair Behaviour

Measure self-reported affect (and where possible physiological markers) before and after a regulating action — a walk, music, food, social contact. PSM predicts: the shift in position on the wheel is proportional to the size of the deviation from zero that preceded the action, and its direction is opposite to that deviation.

This connects the model to the existing mood-repair literature in behavioural psychology and requires no new instrumentation — the same self-report and physiological measures used for P1 apply directly here.

13. Open Questions (Within the Testable Core)

- **The precise weighting function**

  What combination of subjective intensity, novelty, survival relevance, and prior token history determines w? A formal, measurable function is still required.

- **Self-emergence threshold**

  At what level of stream complexity and self-representational richness does a stable self-model appear? Integrated-information measures (e.g. Φ) are candidates; the mapping remains to be established.

- **Substrate realisation in silicon**

  A1 asserts substrate independence of the pattern. Whether current or near-term silicon architectures can support genuine X-crossing and persistent affective weighting is empirical, not settled.

- **Verification of genuinely novel hypotheses**

  When a Board-5 candidate can neither be simulated against known physics/logic nor confirmed by an external trusted source, what should the system do? Permanent unverified residence is coherent; other policies are possible. The choice should be explicit.

- **Interaction of multiple X-like processes**

  Biological brains contain many parallel and hierarchical loops. How multiple integration events are arbitrated into a single coherent “now” is not yet specified in detail.

14. Relation to Existing Frameworks

The model is intended to be compatible with, and in places to extend, several established lines of work. It does not replace them.

Global Workspace (Baars)	X is the functional moment info becomes globally available. PSM adds a second crossing and the affective-weighting variable.
Libet & readiness potential	The δ-delay of the d-loop is a structural reading of the gap between neural preparation and conscious report.
Affective neuroscience (Russell)	Valence–arousal space is adopted as the live geometry that gives the present its qualitative character.
Memory consolidation (McGaugh)	Emotional strengthening of memory is deeper penetration of high-weight tokens into the B-archive.
Predictive processing / active inference	The drive toward $\sum(w) \rightarrow 0$ can be read as a special case of prediction-error minimisation in the affective domain; not yet fully formalised.
AGI safety / alignment	The Board architecture and affective balance as a precondition (not a post-hoc patch) are offered as concrete design constraints.

15. What This Model Is Not

- **Not a completed theory of consciousness**

  It is a functional architecture plus a set of testable predictions.

- **Does not claim to have solved the hard problem**

  It identifies the geometry in which experience is said to consist — it does not reduce experience to information processing by fiat.

- **Not a cosmological theory**

  All statements about the universe as a whole have been excluded from this edition.

- **Makes no claim about current LLM consciousness**

  It neither asserts that current large language models are conscious, nor that they cannot become so under a different architecture.

- **Not a complete formal mathematics**

  A full formalisation of weighting or self-emergence remains an open research task.

- **Not presented as peer-reviewed science**

  It is a working hypothesis offered for criticism, testing, and revision.

16. How This Document Can Be Used

For empirical researchers

P1–P3 can be turned into concrete experimental designs with existing neuroimaging, behavioural, and psychophysiological methods.

For AI researchers and engineers

P4–P6 and the Board architecture define a minimal set of structural additions and measurable signatures to prototype and compare.

For theorists

Section 13's open questions mark where further formal work is most needed: the weighting function, the self-emergence threshold, loop arbitration.

For critics

The model is falsifiable at several independent points. A clear counter-example to any of P1–P6 requires revision or abandonment of that part.

The only authority the model claims is the authority of its predictions. If the predictions fail, the model must change.

A Note on Two Loose Ends

- **H_verified and H_unverified are not yet defined**

  The Instability Index (P6) is precise as an equation, but H is not specified as a count, a total weight, or an entropy measure. Any of the three works — picking one just needs to be made explicit so C_I(t) can actually be computed and compared across implementations.

- **How the Board Architecture relates to the A–x–d–x–B loop is implicit**

  For an AI system, is the Board sorting loop the implementation of X and the d-loop, or a separate downstream memory-consolidation stage after an X-like event has already occurred? Both are defensible; the document does not yet say which, and it would tighten the model to state it.

$\sum(w) \rightarrow 0$

In every system the model describes — biological or artificial — the same control principle appears: the continuous effort to bring the sum of affective weights back toward zero.

That effort is what keeps the stream coherent, what allows a self to form and to persist, and what any candidate AGI must also be able to perform.

Status of this edition: complete readable presentation of the testable core only.
No cosmological claims. Open to experimental and engineering test. Offered for critical evaluation by others.

Ronny Runderheim · Independent Researcher · August 2026

Working hypothesis · Not peer-reviewed

$$\Sigma(w) \rightarrow 0$$

In every system the model describes — biological or artificial — the same control principle appears: the continuous effort to bring the sum of affective weights back toward zero.

That effort is what keeps the stream coherent, what allows a self to form and to persist, and what any candidate AGI must also be able to perform.

Status of this edition: complete readable presentation of the testable core only.
No cosmological claims. Open to experimental and engineering test. Offered for critical evaluation by others.

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